

RESEARCH REPORT · EXAMPLE

Scaling study

What 240 tokenizers taught us

Two algorithms, eight vocabulary sizes and fifteen training mixtures, traced from raw data to every trained vocabulary, then compared with external baselines.

TRAINING COMPLETE · 240 / 240 EVALUATED · 3 BENCHMARKS

2	8	15	264
algorithms	vocabulary sizes	training mixtures	tokenizers evaluated

Created 24 October 2026 · updated 27 October 2026

md2pdf · Brilliant design system

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PART I

The study

What was asked, and how the experiment is built.

1 Summary

2 Design

1 Summary

2026-10-24

240
tokenizers trained
0 failures · 20 h on one laptop

0.589
best boundary F1
Unigram 16K · best baseline 0.492

2.8×
fewer tokens than the baseline
32K, half the data in-language

The study asks one question: **how much vocabulary does a low-resource language need** before extra tokens stop paying for themselves? We trained every combination and measured each one three ways.

NOTE

Every number in this report is traceable to a manifest with a SHA-256 hash; see section 4.

2 Design

2.1 Variables

Factor	Levels	Values
Algorithm	2	BPE, Unigram
Vocabulary size	8	8K → 256K
Mixture	15	0–100% in-language



PART II

Results

How the 240 tokenizers perform against 24 baselines.

3 Unigram beats BPE at every size

4 Reproduce it

3 Unigram beats BPE at every size

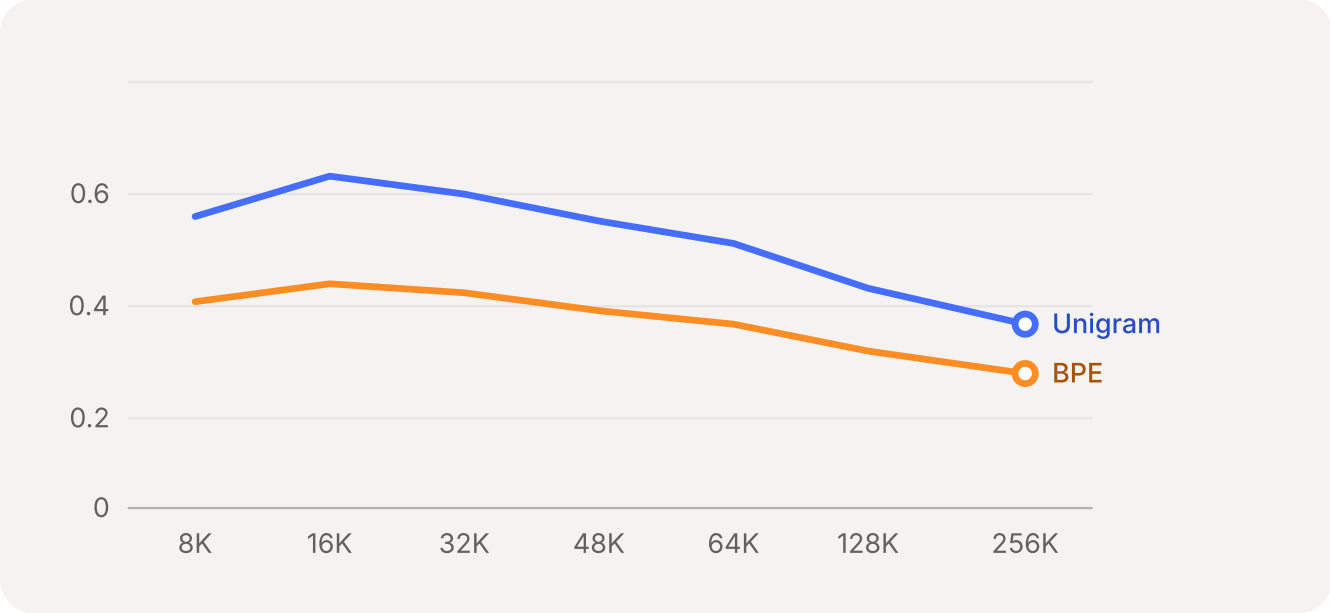


Figure 1. Morpheme-boundary F1 by vocabulary size. Unigram (blue) leads BPE (orange) at every size; both fall after 32K.

The objective Unigram maximises is the likelihood of the corpus under a unigram model:

$$\mathcal{L} = \sum_{s \in D} \log \sum_{x \in S(s)} \prod_{i} p(x_i)$$

CAVEAT

Medians hide the spread: at 256K the worst mixture scores 0.31 lower than the best.

DONE

All 240 models restored from their manifests with 480 / 480 hashes matching.

4 Reproduce it

```python
def train(mix, algo, size):
    # one SentencePiece run per cell of the grid
    return spm.train(input=mix, model_type=algo, vocab_size=size * 1024)  # "deterministic"
```